

## Per Cent of What?

*Four efficiencies wearing the same name, and a writer who forgot they were different*

Read *The Efficiency Cascade: From Starlight to Steak* first. This sheet assumes you have.

**Before you start.** The article you read last lesson made one central claim: that the word *efficiency* is doing four different jobs, and that only one of the four numbers could not have been otherwise. The article below was written by someone who read the same piece and did not notice that.

**Scoring.** +2 for a faulty sentence identified with the right reason. +1 if you spot it but the reason is wrong. -2 for flagging a sound sentence. Maximum eight flags. Several sentences that sound impossible are true, and at least two faults can only be caught by calculating.

### Part A Name the denominator

Every percentage is a percentage *of* something. Fill this in from memory before reading further. This is the whole skill.

| Figure                       | Value | Per cent <i>of what</i> , exactly |
|------------------------------|-------|-----------------------------------|
| Fusion, single event         |       |                                   |
| Fusion, whole star, lifetime |       |                                   |
| Photosynthesis, realised     |       |                                   |
| Trophic transfer             |       |                                   |
| Methane loss in a cow        |       |                                   |
| Cow, gross to net            |       |                                   |

**A1.** Two of these six share a denominator. Which two? \_\_\_\_\_

**A2.** Which single one of the six would be the same in any universe with the same physics? \_\_\_\_\_

### Part B The article

*Nature is wasteful at every scale, and the numbers prove it. A star converts 0.7% of its mass into energy over its lifetime.<sup>[1]</sup> A leaf converts one to two per cent of the sunlight that hits it into stored chemical energy.<sup>[2]</sup> A food chain passes on about ten per cent at each step.<sup>[3]</sup> A cow manages twenty-five to forty per cent.<sup>[4]</sup> Lining them up, the ranking is obvious: cows are the most efficient things in this list, and stars are the least.<sup>[5]</sup> Cows are therefore more efficient than the ecosystems they live in.<sup>[6]</sup>*

*Even the plants beat the stars. Photosynthesis has a theoretical ceiling of 4.6% for C3 plants and 6% for C4 plants such as maize,<sup>[7]</sup> so a field of maize is running at a higher efficiency than the Sun's core.<sup>[8]</sup>*

*The cow's methane losses are four to eight per cent of gross energy,<sup>[9]</sup> which means a cow throws away more energy as methane than a plant ever captures from sunlight.<sup>[10]</sup> Nine-tenths of the Sun's mass never fuses at all, sitting unburned for the whole life of the star.<sup>[11]</sup> If we could persuade the rest of it to fuse, the Sun's lifetime efficiency would rise to 0.7%.<sup>[12]</sup>*

*The reason all four numbers are low is the same in each case: the second law of thermodynamics guarantees that energy is lost as heat at every transformation.<sup>[13]</sup> Evolution has been working on these losses for billions of years and has clearly hit the ceiling in every case.<sup>[14]</sup>*

*Finally, the worksheet's crop-to-human figure of twenty-five per cent must be a misprint, because it is above the trophic range of five to twenty per cent.<sup>[15]</sup> And of the four numbers, only the fusion figure could not in principle have been otherwise.<sup>[16]</sup>*

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| No. | What is wrong, precisely | A sentence you could defend |
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## Part C Do the arithmetic they didn't

**C1.** Sentence 1 and sentence 12 disagree with the article you read last lesson, and with each other. Using the figures from that article, work out the Sun's lifetime efficiency and show the two numbers you multiplied.

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**C2.** Sentence 10 compares four to eight per cent with one to two per cent and concludes that the first is bigger. Both numbers are correct. Explain, in one sentence, why the comparison is still meaningless.

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**C3.** Sentence 6 compares the cow's twenty-five to forty per cent with the ecosystem's ten per cent. Write down what each of those two percentages is measuring, side by side, and then say whether a cow can be said to "beat" the ecosystem at all.

| The cow's 25–40% measures | The ecosystem's 10% measures |
|---------------------------|------------------------------|
|                           |                              |

Verdict:

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## Part D Two students disagree

Both have something right. Say what, and what not.

**Nadia:** “Sentence 13 is fine. The second law really does mean you lose energy as heat every time, and all four numbers are low, so it explains all four.”

**Tom:** “Sentence 13 can’t be right, because the fusion number isn’t about heat loss at all. But then I don’t know what the second law is doing in the article.”

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**Ife:** “Sentence 14 must be true. Evolution has had billions of years, so everything alive must be as efficient as it can be.”

**Callum:** “Sentence 14 is unfalsifiable. Whatever number we measure, you could say evolution hit the ceiling there.”

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## Part E The one that matters

**E1.** Sentence 5 ranks the four numbers against each other. Explain why the ranking is not merely wrong but meaningless, using the word *denominator*.

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**E2.** Rewrite the article’s opening paragraph so that the four figures are stated accurately and no comparison between them is implied.

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**E3.** Last lesson’s article ended by saying only one of the four numbers could not in principle have been otherwise. Sentence 16 repeats that claim and is sound. Say in your own words what makes the fusion figure different in kind from the other three.

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## Teacher key

**Faulty:** 1, 5, 6, 8, 10, 12, 13, 14, 15.    **Sound:** 2, 3, 4, 7, 9, 11, 16.

**1.** The 0.7% is the mass converted *in a single fusion event*, not over a star’s lifetime. Only about a tenth of the Sun’s mass ever reaches the core conditions, so the lifetime figure is  $0.007 \times 0.1 \approx 6.8 \times 10^{-4}$ , roughly 0.07%. A student who read the article last lesson has met both numbers and must notice which one this sentence has taken.

**5 and 6.** The central fault of the sheet. The four figures have four different denominators: mass, incident sunlight, energy in the level below, and gross energy in forage. Ranking them is like ranking a temperature against a speed. Sentence 6 is the same error dressed as ecology, and it is worth noting that the phrase “beating the average” invites exactly this slip even when the underlying article is careful.

**8.** Two errors at once. A stellar core’s energy release is not the same quantity as a leaf’s conversion of incident light, and even taken at face value the comparison sets 6% against 0.7% without asking of what.

**10.** Both figures are correct and the comparison is empty: four to eight per cent *of the gross energy in forage* against one to two per cent *of incident sunlight*. Different denominators, so the larger number is not the larger quantity. This is the subtlest item here and the most transferable beyond the topic.

**12.** Fusing all the Sun’s hydrogen would raise the lifetime figure towards 0.7% only if every nucleus fused, which would also require the star to be something other than a star; more to the point, the sentence treats a hypothetical as an efficiency gain that could be engineered.

**13.** The second law forbids a perfect transformation and fixes no particular number. It also does not apply in the same way to all four: the fusion figure comes from nuclear binding energy, not from heat loss at all, which is the point Tom half-sees in Part D.

**14.** Unfalsifiable as written, and false in detail, since the gap between realised photosynthesis at one to two per cent and the theoretical ceiling at four to six is precisely a case where evolution has *not* reached the ceiling. Callum is right about the logic; reward students who add this concrete counterexample to his point.

**15.** Crop-to-human is not a trophic transfer between two levels of a food web, so the trophic range does not govern it. The reason to reject the sentence is the category, not the number.

**Sound sentences.** **2, 3, 4, 7, 9** are quoted accurately from last lesson’s article and are there to punish flagging by vibe. **11** sounds extraordinary and is true. **16** restates the article’s own conclusion and is the sentence the whole sheet is defending.

**On Part A.** Do this cold, before the article, and collect it before they turn the page. It is the single most diagnostic thing here: a student who cannot say what each percentage is a percentage of will flag sentences 5, 6, 8 and 10 as “sounds odd” at best and miss them at worst. A1: photosynthesis realised and the theoretical ceiling share a denominator (incident sunlight). A2: the fusion single-event figure.

**Sequencing note.** This sheet only works on a second day, and only if last lesson’s article is *not* available during it. Everything sound in Part B is a direct quotation from that piece, so with the article open the task collapses into matching rather than reasoning. Hand the article back afterwards for Part E2.

**If you are running both spot-the-error sheets.** The earlier trophic sheet gives the transfer range as two to twenty-four per cent, from the marine survey literature; this one uses five to twenty, following your article. If students meet both, that discrepancy is worth twenty minutes on its own: two honest sources, two ranges, and the question of what a range is a range *of* (which ecosystems, measured how, over what period). Do not resolve it for them.